Hall Ticket Number:

MR-21

Code: B414IT5S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July/Aug-2024

Discrete Mathematics

(Information Technology)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

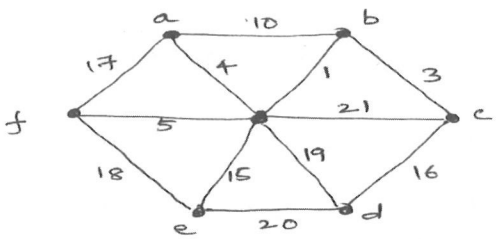
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Translate the following statement into symbolic form i. I shall go to the game or watch the game on television ii. Jack and Jill went up the hill.	2	2	1	1,2
1. b)	What is universal and existential Quantifiers?	2	1	1	1
1. c)	Explain Antisymmetric, irreflexive properties of a relation with examples?	2	2	2	1,2
1. d)	Let X = {1, 2, 3, 4} and R={(x, y)| x > y}. Draw the graph of R and also give its matrix.	2	3	2	1
1. e)	In how many ways can the letters of the word 'MATHEMATICS' be arranged?	2	2	3	1,2
1. f)	How many 3-letter words can be formed using the letters from the word "FABLE"?	2	2	3	1
1. g)	A card is drawn from a well shuffled pack of cards. What is the probability that it is either spade or an ace?	2	4	4	1,2
1. h)	What is a recurrence relation and explain with an example?	2	3	4	1
1. i)	Explain null graph?	2	1	5	1
1. j)	Write Applications of Trees?	2	3	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	$(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ logically equivalent? Justify your answer by using the truth table method.	5	2	1	1
2. b)	Obtain the PDNF for $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$.	5	3	1	1,2
	OR				
2. c)	Show that $P \rightarrow S$ can be derived from the premises $\neg P \vee Q$, $\neg Q \vee R$, and $R \rightarrow S$.	5	3	1	1
2. d)	Establish the validity of the following argument "All integers are rational numbers. Some integers are powers of 2. Therefore, some rational numbers are powers of 2".	5	2	1	1
3. a)	Let X = {1, 2, 3, 4} and R = {(1, 2), (2, 3), (3, 4)} be a relation on X. Find R^*.	5	1	2	1
3. b)	Let R = {(1, 2), (3, 4), (2, 2)} and S = {(4, 2), (2, 5), (3, 1), (1, 3)}. Find $R \circ S$, $S \circ R$, $R \circ (S \circ R)$, $(R \circ S) \circ R$, $R \circ R$ and $S \circ S$.	5	1	2	1
	OR				
3. c)	Let $X = \{1, 2, 3, \ldots, 7\}$ and $R = \{(x, y) \mid x - y$ is divisible by 3. Show that R is an equivalence relation.	5	2	2	1
3. d)	Draw the Hasse diagram representing the positive divisions of 36 (i.e., D36).	5	2	2	1,2
4. a)	Using the principle of mathematical induction, prove that $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \ldots + n(n+1) = (1/3)\{n(n+1)(n+2)\}$.	5	4	3	1,2
4. b)	Find the coefficient of x^5y^2 in the expansion of $(2x-3y)^7$?	5	2	3	1,2
	OR				
4. c)	A committee of 5 people is to be formed from a group of 4 men and 7 women. How many possible committee can be formed if at least 3 women are on the committee?	5	1	3	1,2

P.T.O.

4. d)	State the pigeon hole principle? If there are 5 boxes to keep books find the minimum number of books required to be sure that there are two books in at least one box?	5	2	3	1
5. a)	Two friends Harish and Kalyan appeared for an exam. Let A be the event that Harish is selected and B is the event that Kalyan is selected. The probability of A is 2/5 and that of B is 3/7. Find the probability that both of them are selected.	5	3	4	1,2
5. b)	Solve the recurrence relation $S_n=8S_{n-1} - 16S_{n-2}$ for $n\geq 2$ $S_0=16$, $S_1=80$?	5	3	4	1,2
	OR				
5. c)	Find out the sequence generated by the following function $(3+x)^3$?	5	2	4	1
5. d)	Find the first five terms of the given recurrence relation $a_n=na_{n-1}+n^2a_{n-2}$ where $a_0=1, a_1=1$?	5	3	4	1
6. a)	Define (i) Simple Path (ii) Closed Path (iii) Circuit (iv) Cycle.	5	1	5	1,2
6. b)	Compare and contrast Euler and Hamiltonian graphs using examples?	5	2	5	1
	OR				
6. c)	Find the chromatic number of the k_5 Graph?	5	3	5	1
6. d)	 Define spanning tree. Apply Prim's algorithm to find minimum spanning tree on the following weighted graph.	5	3	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome